

Royal Netherlands Institute for Sea Research

FAIR - UNECSS workshop

Best Practices for Making (Meta)Data FAIR

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Principles of Reusable

- R1. (Meta)data are richly described with a plurality of accurate & relevant attributes
  - R1.1. (Meta)data are released with a clear and accessible data usage license
  - R1.2. (Meta)data are associated with detailed provenance
  - R1.3. (Meta)data meet domain-relevant community standards

R1. (Meta)data are richly described with a plurality of accurate & relevant attributes

- Principle **F2** focuses on having **rich metadata**
- **R1** focuses on assessing if the data are **useful** in a particular context.

- Plurality is used to indicate that the metadata as generous as possible
  - We should not predict the user's identity and needs
  - Write for the unknown other

R1. What should be included?

- Title
- Description
- Co-authors
- Keywords
- Sources
- Data storage paths

But also:

- Experimental protocols
- Date of data generation/collection
- Lab conditions
- Sensor manufacturer/brand
- Species studied
- Samples used
- Support staff involved
- Software/code/apps used
- Databases accessed
- And more!

R1.1. What license can I use?

The most commonly used data licenses are the **Creative Commons (CC) copyright licenses**.

Examples of Creative Commons versions of an open license:

- Public domain: (CC0)
- Attribution: (CC-BY v4.0)
- Attribution & share-alike: (CC-BY-SA v4.0)

The Simpler the license, the better!

R1.3. (Meta)data meet domain-relevant community standards

Reusing datasets becomes easier if they follow:

- Standardized data formats.
- Organized data structures.
- Common metadata templates with consistent vocabulary.

Follow community standards:

Many communities have metadata standards that FAIR data should meet.

Examples: Darwin Core or Ecological Metadata Language (EML).

Try and follow best practices for your field.

- **R1.2 (meta)data are associated with detailed provenance**

What is provenance?

The history of ownership of a valued object or work of art or literature

- **R1.2 (meta)data are associated with detailed provenance**

What is provenance?

Provenance is the documentation that describes where that data/software/sample came from, how it was produced and how it was processed.

R1.2 What is provenance?

Provenance answers the questions on:

- **Who** contributed to the data, who created, and who owns the data?
- **What** processes, materials, protocols, instruments, other data, codes were used?
- **Where** are the samples from, was the work done, are the data/documents?
- **When** was the material collected, data created, data processed?
- **How** was the material and data handled

R1.2 What data needs provenance?

Spreadsheets of observations and measurements

Calculations applied to observations and measurements

Media: images, videos, sound files

Outputs from processing instruments: sequences, microscopes, scanning devices, ...

Outputs from models

Earth Observation data

So pretty much everything….

Sorry ☹

- # R1.2 Why is provenance?

Trustability

I know how your data were produced, I can trust that you did the work and how you did the work

Traceability

I know who did the work and how, that helps me trace the steps followed

Reproducibility

I can repeat what you did and should be able to repeat your results

Reusability

I know what was done to create the data, I know how I can use that data for my own science

R1.2 When is provenance?

All research goes through similar steps

- Data are created
  - **Data (raw) are published**
- Data are processed, transformed
  - **Data (processed) are published**
- Scientific analysis is done
  - **Data (final) are published**



Provenance is relevant for each published dataset

So others can reuse those data with a good scientific confidence

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R1.2 How is provenance?

Start collecting the provenance information from Day 1

You will not remember exactly what and which you used months later

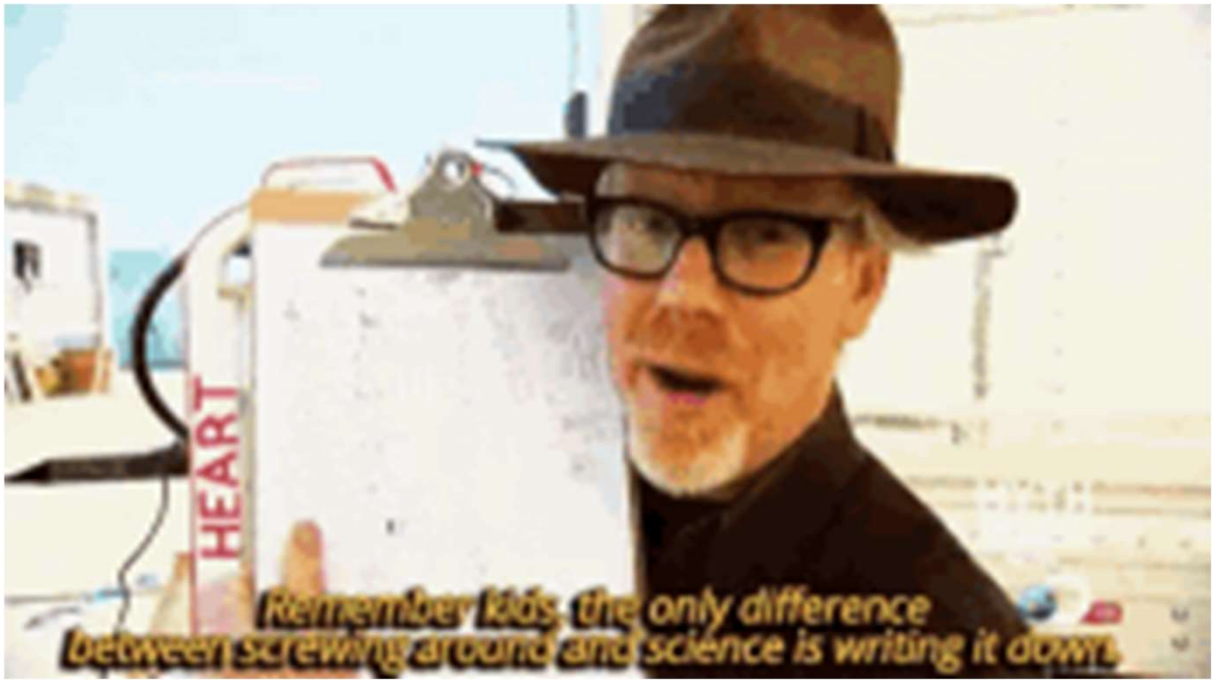
You will spend a boring time (re)gathing this info

Collect the provenance metadata in a specific provenance file

For each research step, list the step and its provenance metadata

Treat provenance like housework, a little every day goes a long way.

- **R1.2 Science is provenance?**



Sea research from climate to biodiversity